THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 12: **Magnetic Effects of Electric Current** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A circular coil carrying current has a south pole facing the viewer. The current as seen is 1

(a) clockwise **(b)** alternating **(c)** zero **(d)** anticlockwise

2. The number of reversals of AC in a minute of 50 Hz supply is 1

(a) 100 **(b)** 3000 **(c)** 600 **(d)** 6000

3. Assertion (A): A conductor moving parallel to the magnetic field induces no current. Reason (R): There is no change in the magnetic field lines linked with the conductor. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

4. The right-hand thumb rule applied to a vertical wire carrying current upward gives the field direction 1

(a) downward **(b)** anticlockwise when seen from above
(c) upward **(d)** clockwise when seen from above

5. A fuse of 5 A is connected in a circuit with a 1.5 kW heater at 220 V. The fuse 1

(a) gives no protection **(b)** does not blow
(c) blows because I = 1.5 A **(d)** blows because I = 6.8 A

6. Assertion (A): AC is preferred for the transmission of power over long distances. Reason (R): Much less energy is lost when AC is sent over long distances. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

7. A positively charged particle moves along the direction of the magnetic field. The force on it is 1

(a) zero **(b)** maximum
(c) along the field **(d)** perpendicular to the field

8. Appliances A (1 kW), B (0.5 kW) and C (0.05 kW) use the same 220 V line protected by a 5 A fuse. Which combination will blow the fuse? 1

(a) A and C together **(b)** B and C together **(c)** A alone **(d)** A and B together

9. **Assertion (A):** Alternating current changes its direction periodically. 1
Reason (R): The frequency of AC supplied in India is 100 Hz.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. **Assertion (A):** A fuse is connected in the neutral wire of a domestic circuit. 1
Reason (R): A fuse protects the appliances by melting when the current is excessive.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. What happens to the induced current if the same magnet is pushed into a coil with a larger number of turns? 2

12. What is the function of the split-ring commutator in an electric motor? 2

13. How would you test whether a wire in a circuit is carrying current without touching it, using a compass? 2

14. Why does a magnet attract an iron nail but not a copper wire? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A heater of 2 kW and a toaster of 1 kW are used at a time on a 5 A fuse circuit at 220 V. Will the fuse blow? Show by calculation. 3

16. How can you determine the polarity of an electromagnet using a compass needle? 3

17. Explain why an alternating current is produced by the coil of a generator when it rotates continuously in the same direction. 3

OR

Explain how a magnet, a coil and a galvanometer can be used to show that the induced current depends on the speed of the magnet. 3

18. Under what circumstances is the force on a moving charge in a magnetic field maximum? 3

OR

How does the output of an AC generator change if the number of turns in the coil is doubled? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study.** In a laboratory, Sneha connects the ends of a coil of insulated copper wire to a galvanometer. When she pushes a bar magnet rapidly into the coil, the needle of the galvanometer deflects momentarily. The needle returns to zero when the magnet is held at rest inside the coil. 5
(i) Name the phenomenon.
(ii) Who discovered it?
(iii) Why does the needle show no deflection when the magnet is stationary?
(iv) State two ways of increasing the induced current.

20. A student has a solenoid, a bar magnet, a galvanometer and a battery. Design activities to show 5
(a) the magnetic effect of current
(b) electromagnetic induction. State the observations and the conclusions.

OR

Describe the effect on the heating and safety of an appliance if 5
(a) the earth wire is not connected
(b) the fuse is placed in the neutral wire
(c) the switch is placed in the neutral wire.

END OF QUESTION PAPER — Total: 40 Marks